MATHS MANTHAN
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Subject: Mathematics **PERIODIC ASSESSMENT TEST – 11 (LINEAR EQUATIONS)** **Time:** 60 Minutes

Class: Class X (CBSE) **Chapter 3: Pair of Linear Equations in Two Variables** **MM: 30**

Student Name: _____ **Roll No.:** _____ **Date:** _____

GENERAL INSTRUCTIONS:

1. All questions are compulsory. Internal choice is provided in Question 14(iv).
2. **Sec A:** 4 MCQs (1M). **Sec B:** 2 AR (1M). **Sec C:** 3 VSAQ (2M). **Sec D:** 3 SA (3M). **Sec E:** 1 LA (5M). **Sec F:** 1 Case Study (4M).

SECTION A: Multiple Choice Questions (4 × 1 = 4 Marks)

Q1. The perimeter of a rectangle is 44 cm and its length is 1 cm more than twice its breadth. The dimensions are: [1]

(A) 15 cm, 7 cm (B) 16 cm, 6 cm (C) 14 cm, 8 cm (D) 18 cm, 4 cm

Q2. If $x = 2, y = 3$ is a solution of the equation $5x - 3y = k$, then the value of k is: [1]
(A) 1 (B) -1 (C) 19 (D) 0

Q3. If $2x + 3y = 11$ and $2x - 4y = -24$, then the value of m for which $y = mx + 3$ is: [1]
(A) 1 (B) -1 (C) 2 (D) -2

Q4. The value of k for which the system $kx - y = 2$ and $6x - 2y = 3$ has a unique solution is: [1]
(A) $k = 3$ (B) $k \neq 3$ (C) $k = 0$ (D) $k \neq 0$

SECTION B: Assertion-Reason Questions (2 × 1 = 2 Marks)

Directions: (A) Both (A) and (R) are true, and (R) is correct explanation. (B) Both true, but (R) is not correct explanation. (C) (A) is true, (R) is false. (D) (A) is false, (R) is true.

Q5. **Assertion (A):** The lines $x = 3$ and $y = -2$ are perpendicular to each other.
Reason (R): The line $x = 3$ is parallel to the y-axis and $y = -2$ is parallel to the x-axis, and the coordinate axes are perpendicular. [1]

Q6. **Assertion (A):** The equations $2x + 3y = 5$ and $4x + 6y = 15$ represent parallel lines.
Reason (R): Two lines $a_1x + b_1y + c_1 = 0$ and $a_2x + b_2y + c_2 = 0$ are parallel if $\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$. [1]

SECTION C: Very Short Answer Questions (3 × 2 = 6 Marks)

Q7. Find the value of k for which the system $2x + 3y = 5$ and $4x + ky = 10$ has infinitely many solutions. [2]

Q8. Solve using elimination: $47x + 31y = 63$ and $31x + 47y = 15$. [2]

Q9. Write a pair of linear equations which has the unique solution $x = -1, y = 3$. How many such pairs exist? [2]

SECTION D: Short Answer Questions (3 × 3 = 9 Marks)

Q10. Four years ago, a mother was four times as old as her daughter. Six years later, the mother will be two and a half times as old as her daughter. Find their present ages. [3]

Q11. Solve the following pair of linear equations graphically: [3]

$$2x + 3y = 12 \quad \text{and} \quad x - y = 1$$

Find the coordinates of the vertices of the triangle formed by these lines and the line $y = 0$.

Q12. If the pair of linear equations $(k+1)x + 3ky + 15 = 0$ and $5x + ky + 5 = 0$ represent coincident lines, find the value of k. [3]

SECTION E: Long Answer / HOTS Question (1 × 5 = 5 Marks)

Q13. Points A and B are 80 km apart from each other on a national highway. A car starts from A and another car starts from B at the same time. If they move in the same direction, they meet in 8 hours, and if they move towards each other, they meet in 1 hour and 20 minutes. Find the constant speeds of the two cars. [5]

SECTION F: Applied Case Study Based Question (1 × 4 = 4 Marks)

Q14. Case Study: Highway Surveillance & Dual Patrol Vehicle Tracking

Two highway police patrol stations, Post Alpha and Post Beta, are positioned along National Highway 44 at a distance of 100 km apart. Highway radar cameras detected two suspect vehicles, Car 1 starting from Post Alpha and Car 2 starting from Post Beta at the same instant.

If both vehicles travel in the same direction along the highway at constant speeds, Car 1 catches up with Car 2 in 5 hours. If they travel towards each other, they pass each other in 1 hour. Let u km/h be the speed of Car 1 and v km/h be the speed of Car 2 ($u > v$).

Based on the above surveillance data, answer the following questions:

(i) Formulate the pair of linear equations in u and v representing the two traffic scenarios. [1]

(ii) Determine the speed of Car 1 (u). [1]

(iii) Determine the speed of Car 2 (v). [1]

(iv) Find the distance traveled by Car 1 before meeting Car 2 when they move towards each other. [1]

OR

Find the ratio of the speed of Car 1 to that of Car 2. [1]

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★★★ END OF QUESTION PAPER · ALL THE BEST ★★★